Lab 11 - Counters and Dividers

In this lab, we learned how to make clock dividers from two types of counters.

Rubric

Item	Description	Value
Summary Answers	Your writings about what you learned in this lab.	25%
Question 1	Your answers to the question	25%
Question 2	Your answers to the question	25%
Question 3	Your answers to the question	25%

Names
Dresden Wood, Alex Worley
Summary

In this lab, we learned how to use counters and clock dividers in Verilog. We created and tested circuits using Vivado and the Basys3 board to see how clock signals can be slowed down and used to control LEDs.

Lab Questions

1 - Why does the Modulo Counter actually divide clocks by 2 * Count?
The Modulo Counter divides the clock by 2 times because the output must toggle twice (LOW to HIGH and HIGH to LOW) to complete one full cycle. For a Count of 6, it takes 12 input clock pulses, resulting in a divide-by-12 clock.

2 - Why does the ring counter's output go to all 1s on the first clock cycle?
The ring counter goes to all 1s on the first clock cycle because it is initialized to that value. This gives the counter a starting point so it can shift the bits through the register on each following clock cycle.

3 - What width of ring counter would you use to get to an output of ~1KHz?

For a ring counter, you would need 100,000 bits to get approximately 1 kHz from a 100 MHz clock. This is because the clock frequency is divided by the number of bits in the ring counter.